



(Data Visualisation & Analytics) Retail Sales Performance Analysis

Final Project Report – Capstone 2

Sector: Retail / E-Commerce

Team: Retail Pirates

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GitHub: https://github.com/ayush-pandey047/Retail_Pirates_SalesAnalysis

Tableau Dashboard: <https://public.tableau.com/app/profile/pranay.chitare/viz/RetailPiratesSalesAnalysis>

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2. Executive Summary

Problem

Retail businesses generate large volumes of transactional data every day, but few are able to use that data to make better decisions. This project addressed a core business challenge: identifying what drives sales performance across product categories, locations, payment channels, and customer segments, and understanding where revenue is being lost or underutilised.

Approach

Transaction data covering over 12,500 records from January 2022 to January 2025 was sourced from a structured retail sales dataset. The data was processed through a three-stage pipeline entirely in Python: extraction, cleaning, and feature engineering. Missing values were imputed, outliers removed via IQR filtering, and nine derived columns were added to enable richer analysis. The cleaned data was then connected to Tableau to build two interactive dashboards — an Executive Summary view and an Operational Drill-Down view.

Key Insights

- Sales performance varies significantly across the eight product categories, with certain categories consistently contributing a disproportionate share of total revenue.
- Discount application was inconsistently recorded — roughly 33% of entries were missing — making it difficult to assess how discounting affects total revenue.
- Customer spending is concentrated in a small segment of high-value buyers, while the majority fall in the low-to-medium spending tier.
- Payment method preferences differ by location, suggesting that channel-specific strategies could improve conversion rates.

Key Recommendations

- Standardise discount tracking across all transactions so that the impact of promotional pricing can be accurately measured and managed.
- Focus retention and upsell efforts on the high-spending customer segment, as this group drives a significant share of total revenue.
- Review stock and supply strategies for top-performing categories to ensure availability, especially in locations with higher footfall or online demand.

3. Sector & Business Context

Sector Overview

The retail sector is one of the largest and most data-intensive industries globally. Retailers collect transaction records at scale, covering product categories, pricing, discounts, payment channels, and geographic locations. The challenge is not in collecting data, but in organising and interpreting it to make decisions that affect revenue, inventory, and customer satisfaction.

Retail businesses face consistent pressure to reduce waste, improve margins, and personalise customer experience. Data analytics has become central to addressing these pressures, particularly for understanding sales patterns and identifying underperforming areas before they become expensive problems.

Decision-Maker

This project is designed for retail operations managers and senior business analysts who need to understand sales trends across categories, regions, and customer types. The dashboards and insights produced are intended to support decisions around pricing, inventory management, and promotional planning.

Why This Problem Was Chosen

Retail sales data is widely available and directly tied to business outcomes. The team chose this problem because it offers a clear path from raw transactional data to actionable business insight. The multi-year dataset spanning 2022 to 2025 also allows for trend analysis across a meaningful time period.

Business Value

Solving this problem provides the business with visibility into where revenue is coming from, which customer segments are most valuable, how discounts are being applied, and which product categories or locations need attention. These insights can reduce revenue leakage, improve inventory

planning, and support targeted marketing decisions.

4. Problem Statement & Objectives

Formal Problem Definition

Retail businesses accumulate large volumes of transaction data without a systematic way to analyse and act on it. Sales performance across categories, locations, customer types, and payment channels is not visible in a format that decision-makers can use. As a result, discount strategies, inventory decisions, and customer engagement efforts are often made without adequate data support.

Project Scope

In scope:

- Analysis of 12,575 retail transactions spanning January 2022 to January 2025
- Data cleaning, feature engineering, and exploratory analysis using Python
- KPI development and dashboard design in Tableau
- Insights and recommendations based on category, location, payment method, and customer segment

Out of scope:

- Real-time or live data integration
- Customer-level CRM data or demographic information beyond what is in the dataset
- Predictive modelling or machine learning

Success Criteria

- A clean, analysis-ready dataset with all key fields populated and validated
- A minimum of eight actionable insights derived from the analysis
- Two functional Tableau dashboards accessible via a public URL
- A set of at least three prioritised business recommendations with estimated impact

5. Data Description

Dataset Source

Source: data/raw/retail_store_sales.csv (committed to GitHub repository) Direct Access:
https://github.com/ayush-pandey047/Retail_Pirates_SalesAnalysis/tree/main/data

Data Structure

Attribute	Detail
Total Records	12,575 transactions
Time Period	January 2022 - January 2025
Customers	25 unique customers

Product Categories	8 categories

Payment Methods	3 (Cash, Credit Card, Digital Wallet)
Purchase Channels	2 (Online, In Store)

Column-by-Column Explanation

Column	Description
Transaction ID	Unique identifier for each transaction
Customer ID	Identifier for the customer making the purchase
Category	Product category (e.g., Electronics, Clothing, Grocery)
Item	Specific product purchased
Price Per Unit	Selling price of a single unit in the transaction
Quantity	Number of units purchased
Total Spent	Total amount paid by the customer (Price Per Unit x Quantity, post discount)
Payment Method	Mode of payment: Cash, Credit Card, or Digital Wallet
Location	Store or delivery location associated with the transaction
Transaction Date	Date on which the transaction occurred
Discount Applied	Whether a discount was applied to the transaction (Yes/No/Missing)

Data Limitations and Known Biases

- Approximately 33% of Discount Applied values (4,199 out of 12,575) were missing, limiting discount impact analysis.
- Only 25 unique customers are present, which is unusually low for over 12,500 transactions. This suggests a simulation or anonymised dataset rather than a true population sample.
- No demographic data such as age, gender, or purchase history is included, limiting customer profiling.
- The dataset does not include cost-of-goods data, so profit margin analysis is not possible.

6. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were carried out in Python using Jupyter Notebooks. The notebooks are committed to the GitHub repository under the notebooks/ directory.

Stage 1: Extraction (notebooks/01_extraction.ipynb)

- Loaded raw CSV from data/raw/retail_store_sales.csv using Pandas
- Inspected shape, column names, data types, memory usage, and descriptive statistics
- Identified missing values, duplicates, and unique value counts per column

Stage 2: Cleaning (notebooks/O2_cleaning.ipynb)

Duplicate Removal Duplicate rows were identified and removed to ensure each transaction is counted once.

Missing Value Treatment

- Numeric columns: missing values imputed using column median
- Categorical columns: missing values imputed using column mode
- Discount Applied: imputed using mode given ~33% missing rate

Outlier Detection and Removal Outliers were detected using the Interquartile Range (IQR) method on numeric fields. A total of 157 rows were removed as outliers.

Data Type Conversion The Transaction Date column was converted from string to datetime format to enable time-based analysis.

Column Name Standardisation All column names were converted to lowercase with underscores to ensure consistency across notebooks.

Cleaned data was exported to: data/processed/cleaned_data.csv

Stage 3: ETL & Feature Engineering (notebooks/O3_etl.ipynb)

Nine new columns were derived from the cleaned data:

Derived Column	Logic
year	Extracted from transaction_date
month	Extracted from transaction_date
quarter	Extracted from transaction_date
day_of_week	Extracted from transaction_date (0 - Monday)
revenue_per_unit	total_spent / quantity
is_bulk	Flag 1 if quantity above a defined threshold, else 0
spent_segment	Categorised as Low / Medium / High based on total_spent quartiles
discount_flag	Binary: 1 if discount applied, 0 otherwise
category_code	Numeric encoding of product category

Assumptions During Cleaning

- Rows with extreme values in price_per_unit and quantity were treated as outliers rather than genuine high-value transactions.
- Mode imputation for Discount Applied assumes that the most common discount status is representative of missing cases.

7. KPI & Metric Framework

KPI	Definition	Business Relevance
Total Revenue	Sum of total_spent across all transactions	Measures overall sales performance
Average Order Value (AOV)	Mean of total_spent per transaction	Indicates typical customer spend per visit
Revenue by Category	total_spent grouped by category	Identifies best and worst performing product lines
Revenue by Location	total_spent grouped by location	Highlights high-performing regions or stores
Discount Rate	Percentage of transactions with discount applied	Tracks promotional activity and potential margin impact
Payment Method Split	Count and revenue share by payment method	Informs payment channel strategy
Spending Segment Distribution	Count of customers in Low / Medium / High segments	Supports customer tiering and targeting
Monthly Revenue Trend	total_spent aggregated by year and month	Tracks growth or decline over time
Bulk Purchase Rate	Percentage of transactions flagged as is_bulk	Identifies bulk-buying behaviour patterns

All KPIs are computed in notebooks/03_etl.ipynb and notebooks/04_eda.ipynb using Pandas aggregations and are referenced directly in Tableau using the exported cleaned dataset.

8. Exploratory Data Analysis

All EDA was conducted in notebooks/04_eda.ipynb. Charts and summary tables are committed to the GitHub repository.

Trend Analysis — Revenue Over Time

Monthly revenue was aggregated from 2022 through 2025 to observe growth or decline patterns. The data spans all 12 months across three full years, enabling seasonal comparison. Any months showing significant dips or spikes were noted for further investigation in the statistical analysis phase.

Comparison Analysis — Category and Location Performance

Revenue was broken down by the eight product categories and by location. This comparison identifies which categories drive the most revenue and which are underperforming relative to their transaction volume. Location-level comparison reveals whether performance differences are structural or driven by local demand.

Distribution Analysis — Customer Spending

The `spent_segment` column, derived during feature engineering, was used to analyse the distribution of customers across Low, Medium, and High spending tiers. A large concentration in the Low tier suggests that average transaction size has room to grow through upselling or bundle strategies.

Payment Method Analysis

Transaction counts and revenue were split by Cash, Credit Card, and Digital Wallet. Understanding payment method preference by location or category helps tailor payment infrastructure and promotional offers.

Discount Impact

Transactions with and without discounts were compared on `total_spent` and quantity. Given the high rate of missing discount data, findings here are directional rather than conclusive. Initial analysis suggests that discounted transactions did not always result in higher quantities, indicating potential misalignment between discount strategy and customer response.

Correlation Analysis

Correlations between quantity, `price_per_unit`, `total_spent`, and derived fields were computed. A positive correlation between quantity and `total_spent` was confirmed. Price per unit and quantity showed a weak negative correlation, consistent with expected consumer behaviour where higher-priced items are purchased in smaller quantities.

9. Statistical Analysis

Statistical analysis was carried out in notebooks/`04_statistical_analysis.ipynb`.

Segmentation

Customers were segmented into three spending tiers — Low, Medium, and High — based on their `total_spent` values. The High segment, while smallest by count, accounts for a disproportionately large share of total revenue, confirming the importance of retaining these customers.

Anomaly Detection

The IQR-based outlier removal conducted during cleaning identified 157 anomalous transactions. These records were removed from the analysis dataset but logged separately for review. Some may represent genuine bulk corporate purchases rather than data errors, which is an area for future investigation.

Seasonal and Quarterly Patterns

Revenue was aggregated by quarter to test for seasonal patterns. Quarters showing higher revenue were examined against payment method and category data to identify whether specific product types drive seasonal peaks.

Day-of-Week Analysis

The `day_of_week` derived column was used to examine whether certain days of the week see higher transaction volumes or revenue. This informs staffing, promotional scheduling, and online campaign timing.

10. Tableau Dashboard Design

Dashboard Objective

The Tableau dashboards are designed to help operations managers and business analysts monitor sales performance, identify trends, and drill into category or location-level details without needing to run queries or code.

Dashboard 1: Executive Summary

URL:

<https://public.tableau.com/app/profile/pranay.chitare/viz/RetailPiratesSalesAnalysis/ExecutiveSummary>

This view provides a high-level snapshot of overall business performance. It includes total revenue, average order value, revenue by category, and monthly trend lines. It is designed for senior stakeholders who need a quick read of how the business is performing across its key dimensions.

Dashboard 2: Operational Drill-Down

URL:

<https://public.tableau.com/app/profile/pranay.chitare/viz/RetailPiratesSalesAnalysis/OperationalDrill-Down>

This view allows analysts to filter by location, category, payment method, and time period. It supports investigation of specific performance questions such as which locations underperform in a given quarter or which categories attract the highest-value customers.

Interactive Elements

- Category filter: allows selection of one or more product categories
- Location filter: filters all views by geographic region or store
- Date range selector: restricts the view to a defined time period
- Payment method toggle: isolates transactions by Cash, Credit Card, or Digital Wallet

GitHub References

Dashboard screenshots are stored in `tableau/screenshots/` and dashboard links are documented in `tableau/dashboard_links.md`, both committed to the GitHub repository.

11. Insights Summary

1. A small number of product categories generate a majority of revenue, meaning the business is disproportionately dependent on a few lines. A disruption in those categories — whether from stockouts or pricing changes — would have an outsized impact on overall performance.
2. The Discount Applied field was missing in approximately 33% of records. This means the business currently has no reliable way to measure whether its discounting strategy is working, and decisions are being made without a clear picture of promotional impact.
3. High-spending customers represent a small fraction of the total customer base but account for a large share of revenue. Losing even a few of these customers would meaningfully affect total revenue.

4. Payment method usage differs by location, suggesting that some stores or online channels are not offering the payment options that customers in those areas prefer. This could be contributing to abandoned purchases or lower conversion rates.
5. Certain days of the week consistently show higher transaction volumes. If staffing, stock, and promotions are not aligned with these peaks, the business is likely leaving revenue on the table.
6. The bulk purchase flag analysis shows that is_ bulk transactions occur regularly but represent a small share of total records. These bulk buyers typically have higher total spend per transaction, making them a valuable but under-served segment.
7. Revenue per unit varies significantly across categories, indicating that pricing strategy is not uniform. Categories with low revenue per unit but high transaction volumes represent opportunities for price optimisation.
8. Quarterly revenue analysis shows variability across periods, suggesting that the business has seasonal exposure. Without a structured response to slower quarters, revenue gaps go unaddressed.
9. The low-to-medium spending segment makes up the majority of transactions. Even a modest increase in average order value within this segment — through bundling or upselling — would produce measurable revenue gains across the dataset.
10. The three-year span of the dataset provides enough history to detect trends, but the small number of distinct customers (25) limits how confidently these trends can be generalised to a larger retail population.

12. Recommendations

#	Insight	Recommendation	Expected Business Impact	Feasibility
1	Discount data is 33% missing	Implement a mandatory discount flag at the point of sale so that every transaction captures whether a discount was applied	Enables accurate ROI measurement on promotions; prevents revenue leakage from unmeasured discounting	Low cost — requires a system-level change to POS or data entry workflow
2	High-spending customers drive outsized revenue share	Create a loyalty or retention programme targeted at the High-spending segment, with personalised offers and early access to new products	Retaining even a portion of this segment can protect a significant share of annual revenue	Medium effort — requires customer identification and a basic CRM capability
3	Top categories carry disproportionate revenue risk	Diversify the product mix and ensure stock levels for top-performing categories are prioritised in supply planning	Reduces revenue volatility and ensures that high-demand products are consistently available	Medium effort — requires coordination between buying, operations, and supply chain teams
4	Payment method preferences differ by location	Audit payment options available at each location and align them with the most-used methods in that area	Reduces friction at checkout and may recover abandoned purchases or improve conversion	Low effort — primarily a configuration and infrastructure review task
5	Revenue per unit varies significantly across categories	Review pricing strategy for low revenue per unit, high volume categories to identify whether price increases are feasible without significant volume loss	Even a 5% improvement in revenue per unit across high volume categories would have a noticeable revenue impact	Medium effort — requires pricing analysis and potentially market testing

13. Impact Estimation

Recommendation 1: Fix Discount Data Capture

Currently, 33% of discount records are missing. Closing this gap does not directly generate revenue but removes a blind spot. If even 10% of discounts being applied were unauthorised or excessive, correcting them could recover that portion of margin. For a dataset of this scale, that represents a non-trivial improvement in promotional efficiency.

Recommendation 2: High-Spend Customer Retention

If the High spending segment accounts for approximately 30–40% of total revenue (a conservative estimate based on typical retail Pareto distributions), a 10% improvement in retention within that segment would protect 3–4% of total annual revenue. For a business with significant transaction volume, this is material.

Recommendation 3: Category Supply Prioritisation

Stockouts in top-performing categories result in lost sales. If top categories account for 50% of revenue and even 2% of their transactions are lost to unavailability, restoring that availability recovers 1% of total revenue — meaningful at scale with low incremental cost.

Recommendation 4: Payment Method Alignment

If 5–10% of potential transactions in certain locations are abandoned due to unavailable payment methods, adding those options could recover that revenue directly. The implementation cost is largely one-time.

Recommendation 5: Pricing Review for High-Volume Categories

A 5% increase in revenue per unit across high-volume, low-margin categories, if demand holds, directly adds 5% to the revenue from those lines. The key assumption is that demand is inelastic enough to absorb the change.

Note: These estimates are illustrative. The logic is directionally sound based on the data, but precise figures require actual cost-of-goods, margin data, and customer elasticity information not present in this dataset.

14. Limitations

- The dataset contains only 25 unique customers across 12,575 transactions. This is not representative of a real retail population and significantly limits the generalisability of customer-level insights.
- Approximately 33% of Discount Applied values were missing. Mode imputation was used, but this introduces uncertainty in any analysis involving discount behaviour.
- The dataset does not include cost-of-goods or profit margin data. All analysis is revenue-based, and conclusions about profitability cannot be drawn.
- No customer demographic data (age, gender, income level) is available, which limits the depth of customer segmentation.
- The 157 rows removed as outliers during IQR filtering may include genuine high-value transactions. This is an assumption that cannot be verified without additional context.
- Analysis is limited to the period January 2022 to January 2025. Conditions before or after this period are not reflected.

15. Future Scope

- Integrating cost-of-goods data would enable profit margin analysis, which is more actionable than revenue analysis alone.
- Adding customer demographic data would support proper customer segmentation and personalisation strategies.

- A time-series forecasting model (such as ARIMA or Prophet) could be built on the monthly revenue data to project future performance and support inventory planning.
- A real-time Tableau dashboard connected to a live database or API feed would allow managers to monitor sales as they happen, rather than reviewing historical data.
- Expanding the dataset to include returns, cancellations, and customer complaints would give a more complete picture of the customer experience and its impact on revenue.
- A proper A/B test on discount application — where some customers receive discounts and others do not — would provide causal evidence of discount impact, rather than the correlational analysis that is currently possible.

16. Conclusion

This project analysed over 12,500 retail transactions spanning three years, cleaning and transforming the data in Python before building interactive dashboards in Tableau. The analysis revealed that revenue is concentrated in a small number of product categories and a small segment of high-value customers, that discount tracking has significant gaps, and that payment preferences differ by location. Acting on the recommendations — particularly around discount data capture, high-spend customer retention, and category supply prioritisation — would give the business clearer visibility into its performance and concrete levers to improve it. The work demonstrates how a structured analytics pipeline can turn raw transactional data into decisions that protect and grow revenue.

17. Appendix

A. Data Dictionary

Column Name	Type	Description	Example Value
transaction_id	String	Unique transaction identifier	TXN-00001
customer_id	String	Customer identifier	CLST-005
category	String	Product category	Electronics
item	String	Specific item purchased	Laptop
price_per_unit	Float	Unit selling price	499.99
quantity	Integer	Units purchased	2
total_spent	Float	Total transaction value	999.98
payment_method	String	Cash / Credit Card / Digital Wallet	Credit Card
location	String	Store or region	North
transaction_date	Date	Date of transaction	2023-06-15
discount_applied	String	Yes / No / Missing	Yes
year	Integer	Year extracted from transaction_date	2023
month	Integer	Month extracted from transaction_date	6

quarter	Integer	Quarter extracted from transaction_date	2

day_of_week	Integer	Day of week (0-Monday)	2
revenue_per_unit	Float	total_spent / quantity	499.99
is_bulk	Integer	1 if bulk purchase, 0 otherwise	0
spent_segment	String	Low / Medium / High	High
discount_flag	Integer	1 if discount applied, 0 otherwise	1
category_code	Integer	Numeric encoding of category	3

B. Notebook References

- notebooks/01_extraction.ipynb — Data loading and initial inspection
- notebooks/02_cleaning.ipynb — Missing value treatment, duplicate removal, outlier filtering
- notebooks/03_etl.ipynb — Feature engineering and final dataset export
- notebooks/04_eda.ipynb — Exploratory data analysis and visualisations
- notebooks/04_statistical_analysis.ipynb — Segmentation and statistical tests

18. Contribution Matrix

This section must clearly document each team member's contribution across all project phases. Claims must match evidence in GitHub Insights, PR history, and committed files. Any mismatch may result in individual grade adjustments.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard
Ayush Kumar Pandey	Primary sourcing, GitHub setup	Support	Support	Contributed	Contributed
Harshit Gupta	Contributed	Support	Core Insights	Lead	Contributed
Pranay Chitare	Dataset handling	Support		Support	Lead
Soham Goel	Lead (Data collection)	Support	Evaluation	Support	Contributed
Ankit Kumar	Contributed	Lead (ETL Pipeline)	Support	Support	Contributed